DATA VISUALIZATION & ANALYTICS

DVA CAPSTONE 2 | FINAL PROJECT REPORT

US Flight Operations

A Plain-English Report on Flight Delays in 2015

US Aviation · Year of Data: 2015

We looked at 100,000 flights from 2015 to figure out when delays happen, why they happen, and what airlines and airports could do to fix them. We used Python to clean the data and Tableau to build the dashboard.

Project Title	US Flight Operations — A Plain-English Look at 2015 Delays
Sector	Aviation / Transportation
Team	Section D — Group 19
Institute	Newton School of Technology
Program	Data Visualization & Analytics — Capstone 2
Repository	github.com/adamyatiwari12/SectionD_G19_FlightOps-Dashboard

TEAM GROUP-19 — SECTION D

Role	Member
Project Lead	Adamya Tiwari
Data Lead	Prateek
ETL Lead	Yash Raghubanshi
Analysis & Lead	Rahul
Visualization Lead	Agnik Misra

Strategy Lead	Rahul
PPT & Quality Lead	Ram

Executive Summary

Flight delays cost the US economy about $32 billion every single year. That's money lost by passengers stuck at airports, by airlines paying extra crew and fuel, and by airports running over capacity. We wanted to find out where this money is being lost and what could be done to save it.

We pulled official flight records from the US government's Bureau of Transportation Statistics for the year 2015 and took a sample of 100,000 flights to study. We cleaned the data, looked at it from many angles, ran some statistical tests to make sure our findings were real, and built an interactive Tableau dashboard so anyone can explore it. Here is what we found.

> *Our main question: Where in the US flight system can we make small changes that lead to big improvements? Which airlines, airports, and times of day deserve attention first?*

The Numbers At A Glance (from our 100,000-flight sample)

TOTAL FLIGHTS	ON-TIME RATE	AVG DELAY	CANCELLED
100K	**81.8%**	**4.4 min**	**1.55%**
from 2015	*arrived within 15 min*	*small, but with a long tail*	*about 1 in 65 flights*

What We Learned (the four big findings)

- If a flight leaves late, it almost always lands late. The connection is so strong that fixing departures alone would solve about 89% of the arrival-delay problem.
- Most delays come from the same plane being late on its previous trip — a domino effect. Weather causes a lot of cancellations, but only about 3% of delay minutes.
- Airlines are not equal. The best airline (Alaska) lands an average of 1 minute early. The worst (Frontier) lands 13 minutes late. That's a 14-minute difference — every flight, every day.
- Mornings are the best time to fly. Morning flights run an average of 3.5 minutes late. Evening flights run 14 minutes late — four times worse for the same airline, same airport.

What Airlines and Airports Should Do (our top three recommendations)

- Schedule important and connecting flights in the morning. They are simply more reliable.
- Fix the turnaround process at the worst airports — Chicago O'Hare, Newark, and San Francisco. These three keep showing up in the slowest-route lists.

- Get extra crew and standby planes ready for summer (June to August) and the winter holidays (December to February). The data shows these are the worst delay periods every year.

We built this whole project using Python (for cleaning and statistics) and Tableau (for the dashboard). All the code, notebooks, and the dashboard are public on GitHub.

1. Why This Problem Matters

1.1 Why Flight Delays Are A Big Deal

Air travel is one of the most time-sensitive industries in the world. Every year, US airlines fly about 800 million passengers on over 9 million flights. When even one of those flights runs late, the effect spreads: the same plane is late for its next trip, the next crew has to wait, passengers miss their connections. By lunchtime, a single delay at 7 AM can have caused six more delays across the country.

The numbers are real. The FAA estimates that more than one in four US flights misses its scheduled arrival time. Every minute of delay costs about $74 — that's fuel, crew overtime, customer compensation, and rebooking fees. Multiply that across millions of flights and you get the $32 billion annual figure.

1.2 Who Would Use Our Dashboard

We built our dashboard with three kinds of people in mind:

- Airline operations managers — they need to know which routes and times need extra buffer, and where to add capacity.
- Airport operations directors — they need to find the slow spots in taxiing and turnaround that ripple out into the whole network.
- Government regulators — they need to spot airlines that cancel flights at unusually high rates.

1.3 What's At Stake

Industry studies say each minute of mainline domestic flight delay costs about $74.20. Even a 5% reduction in average delay across our 100,000-flight sample would save around $1.6 million. Scale that up to the full US flight year (5.8 million flights) and you're talking about hundreds of millions of dollars in savings.

2. The Question We Tried To Answer

2.1 The Question, In One Sentence

Using 2015 US flight data, find out which airlines, airports, times, and patterns cause the most delays — and figure out what would make the biggest difference if we fixed it.

2.2 What We Looked At, And What We Didn't

We did look at:

- All US domestic flights from 2015
- 14 airlines (mainline plus regional)
- 322 airports across the country
- Delays broken down by airline, airport, route, time of day, and season

We did not look at:

- International flights or cargo flights
- Real-time prediction (we only looked back at history)
- Ticket prices, frequent flyer data, or revenue

2.3 How We Decided If We Succeeded

Goal	What It Means	Did We Hit It?
Reproducible code	Anyone can re-run our work with one command	Yes
Real statistics	Every big claim is backed by a statistical test	7 tests run
Useful dashboard	At least 4 dashboard views with filters	Done
Real recommendations	Each recommendation is tied to something we measured	5 written
Public and auditable	All code and notebooks on GitHub	Yes

3. Where The Data Came From

3.1 The Source

We used the official 2015 On-Time Performance dataset from the US Department of Transportation's Bureau of Transportation Statistics. It's a public dataset that records every commercial domestic flight in America. The full file has 5.83 million flights across 14 airlines and 322 airports.

We took a smaller sample of 100,000 flights from this data. We did this so the notebooks would run quickly on a regular laptop. The sample is stratified — meaning it covers all months, all airlines, and all major airports proportionally, so nothing important is missing.

3.2 What's In The Data

File	Rows	Columns	What It Covers
airlines.csv	14	2	Airline names + codes
airports.csv	322	7	Airports + locations
flights.csv (sample)	100,000	31 raw → 61 cleaned	Every flight in our sample

3.3 The Important Columns (in plain English)

Column	What It Is
AIRLINE	Two-letter code for the airline (AA = American, DL = Delta, etc.)
ORIGIN_AIRPORT / DESTINATION	Three-letter airport codes (LAX, JFK, ATL, etc.)
DEPARTURE_DELAY	How many minutes late the plane left the gate
ARRIVAL_DELAY	How many minutes late the plane reached the destination — our main number
TAXI_OUT / TAXI_IN	Time spent rolling on the ground at each end
DISTANCE	How far the flight went, in miles
CANCELLED / DIVERTED	Whether the flight was cancelled or sent somewhere else
AIR_SYSTEM / WEATHER / AIRLINE delays	Why the flight was late (broken into 5 reasons)
CANCELLATION_REASON	If cancelled: was it weather, the airline, security, or air traffic?

3.4 What The Data Can't Tell Us

- Only one year (2015) — we can't see how things changed over time.
- The reason for each delay is reported by the airline itself — they have a reason to call things 'weather' rather than 'our fault'.
- About 8% of the records use old airport codes that we couldn't match to city names — so those rows show up without locations.
- Our 100,000-flight sample is fine for big patterns but might miss very rare events (like security cancellations, which are less than 0.04% of records).

4. Cleaning Up The Data

Real data is messy. Before we could analyze anything, we had to clean it up and add some helpful columns. We did everything in Python (using the pandas and numpy libraries) inside Jupyter notebooks. The whole pipeline runs with one command on GitHub.

4.1 The Pipeline At A Glance

Raw data → Look at it (notebook 01) → Clean it up (notebook 02) → Explore patterns (notebook 03) → Run statistics (notebook 04) → Save final files for Tableau (notebook 05)

4.2 What We Did, Step By Step

Step	Problem We Found	What We Did About It
1	About 81,646 missing values in the 5 delay-cause columns	Filled them with 0 — because if a flight had no weather delay, the value should be 0, not blank
2	98,446 missing values in the cancellation reason column	Filled them with 'Not Cancelled', because that's what missing actually meant
3	260 missing tail numbers	Filled them with 'UNKNOWN'
4	Date stored as separate year/month/day columns	Combined them into one proper date column
5	Times stored as numbers like 1430 (instead of 14:30)	Converted them all to readable HH:MM format
6	Memory usage was 43 MB	Converted text columns to 'category' type — got it down to 38 MB

7	No easy way to slice by time of day	Added a TIME_OF_DAY column (Morning, Afternoon, Evening, Night)
8	Hard to compare delay severity	Added a DELAY_CATEGORY column (Early, Minor, Moderate, Severe, Cancelled)
9	No single column for the main delay reason	Added PRIMARY_DELAY_CAUSE — picks the biggest of the 5 delay-type columns
10	No easy way to group flights by route	Added a ROUTE column (e.g., 'LAX → JFK')
11	Airline and airport codes only — no full names	Joined in airline names, airport names, cities, states, and lat/long

4.3 The Decisions We Made (and why)

- We filled missing delay-reason values with 0 instead of dropping those rows. Reason: in this dataset, missing means 'no delay of that type', not 'unknown'. Dropping would have lost 80% of our data.
- We kept extreme outliers (flights delayed by more than 3 hours). Reason: those are real events, not data errors, and they are exactly the cases that hurt passengers the most.
- We kept the rows with old-style airport codes even though we couldn't look up their cities. Reason: dropping them would have biased our airport rankings.

All the cleaning code is in our notebooks on GitHub — 01_extraction.ipynb and 02_cleaning.ipynb.

5. The Numbers We Care About (KPIs)

Before we started analyzing, we picked six numbers that would tell us if a flight or an airline was performing well. These are our KPIs (Key Performance Indicators) — basically, the report-card numbers.

KPI	How We Calculated It	Why It Matters	Industry Target
On-Time Rate	How often a flight lands within 15 min of schedule	The basic 'are flights running on time?' number	Above 80%
Average Arrival Delay	Average minutes late, across all flights	Quick way to compare months or airlines	Below 5 min
Cancellation Rate	Cancelled flights as a percentage of total	How reliable is the schedule?	Below 2%

Severity Mix	How many flights were a little late vs. very late	Catches the worst delays that the average hides	Severe < 1%
Cascade Strength	How tightly departure delay predicts arrival delay	Tells us if the cascade is the main problem	Track it
Main Delay Reason	Which of the 5 reasons shows up the most	Tells airlines where to spend money fixing things	Track it

All the calculation code is in notebook 05_final_load_prep.ipynb. We saved the results as 7 small CSV files that Tableau can load quickly.

6. What We Found When We Looked At The Data

This section is the fun part. We explored the data from every angle — by month, by airline, by hour of the day, by route — and wrote down what stood out. Each chart is in our notebook 03_eda.ipynb.

6.1 The Big Picture

TOTAL FLIGHTS	ON-TIME	AVG DEPARTURE	AVG ARRIVAL
100K	**81.8%**	**9.4 min**	**4.4 min**
2015 sample	*≤ 15 min late*	*median: -2 min*	*median: -5 min*

6.2 Delays Change A Lot By Month

Looking month by month, we saw the average delay swing from -0.4 minutes (October — flights actually arrived early on average) to 9 minutes (June). Cancellations also peaked in February at 4.4% — that's about 11 times worse than September. The pattern is clear: summer (June through August) and winter holidays (December through February) are the worst times.

> *What this means: if airlines staffed up extra crews in summer and winter holidays — instead of using the same staff levels year-round — they would handle peak demand much better.*

6.3 Day Of The Week Matters Too

Saturdays had the lowest delay rates. Thursdays and Fridays had the highest. This makes sense: business travelers fly more at the end of the work week, so the system gets stressed. The number of flights is roughly the same each day — so it's not about volume, it's about who's flying and where they're going.

6.4 Mornings Are Magic

This is one of our most important findings. Average arrival delay grows all day long:

Time Of Day	Avg Arrival Delay	What It Means
Morning (5 AM - 11 AM)	3.5 min	Best window for connections
Afternoon (12 PM - 4 PM)	4.8 min	Cascade starting
Evening (5 PM - 8 PM)	11.7 min	Earlier delays piling up
Night (after 9 PM)	14.0 min	Worst time of day

6.5 Some Airlines Are Just Better

Average delay by airline ranges from -1.1 minutes (Alaska Airlines arrives early) to +13.3 minutes (Frontier Airlines arrives late). That's a 14.4-minute spread between the best and worst airlines. We tested this statistically and it's a real difference, not just chance. Hawaiian and Alaska are the punctuality champions; United and Spirit are the laggards.

6.6 Some Airports Are Worse Than Others

Atlanta (ATL) handles the most flights of any airport in our sample (about 31,000) but still keeps delays moderate — that's impressive. Chicago O'Hare (ORD), Newark (EWR), and San Francisco (SFO), on the other hand, keep showing up in our most-delayed-route lists. They're crowded hubs that need help.

6.7 Why Flights Are Late

Reason	Share of Delayed Flights	What Could Help
Late Aircraft (cascade from earlier flights)	38.4%	Faster turnaround at the gate
Air System (air-traffic control)	29.0%	Better slot coordination
Airline (crew, fleet, ground ops)	29.0%	Internal airline fixes
Weather	3.4%	Pre-position resources for storms
Security	0.2%	TSA process — minor issue

6.8 Why Flights Get Cancelled

Out of 1,554 cancelled flights in our sample: 52% were because of weather, 28% were the airline's fault, 19% were national-airspace issues. Weather cancels the most flights, but our statistics show that airlines also matter a lot — some airlines cancel at three times the rate of others. That's something regulators should look into.

6.9 Distance Doesn't Really Matter

We expected long flights to have more delays. They don't. Short flights (under 500 miles), medium flights (500-1500 miles), and long flights (over 1500 miles) all have similar delay rates. This means the problem is

happening on the ground (taxiing, turnaround) — not in the air. That's good news, because ground operations are easier to fix than weather.

6.10 The Long Tail of Bad Delays

Most flights are on time or early (median delay is -5 minutes — flights arrive five minutes early on average). But about 0.8% of flights are delayed by more than 3 hours. These rare, extreme delays cause most of the passenger complaints, missed connections, and rebooking costs.

6.11 What's Connected To What

Two Things We Compared	Connection Strength (r)	Plain Reading
Departure Delay ↔ Arrival Delay	0.94	Almost perfectly linked — the cascade
Late Aircraft Delay ↔ Total Delay	0.65	Strong connection
Airline Delay ↔ Total Delay	0.57	Strong connection
Distance ↔ Air Time	0.99	Obvious — sanity check
Day Of Week ↔ Arrival Delay	0.02	No real connection

About the 'r' number: it's how tightly two things move together. r=0 means no connection at all. r=1 means they move together perfectly. Our 0.94 between departure and arrival delay is about as strong as it gets in real data.

7. Proving Our Findings With Statistics

Looking at charts can be misleading. Our eyes find patterns in random noise. So we ran proper statistical tests on every big claim. Here's what we did and what came back.

7.1 The Tests We Ran

What We Tested	Test Name	Result	Verdict
Do airlines really differ in delays?	ANOVA	$F = 39$, $p \approx 0$	Yes, real difference
Is cancellation rate tied to airline?	Chi-Squared	$\chi^2 = 693$, $p \approx 0$	Yes, real link
Are dep delay and arr delay connected?	Pearson r	$r = 0.944$	Yes, very strong
Can we predict arrival delay?	Multiple Regression	$R^2 = 0.943$	Yes, very accurately

Does time of day matter?	Kruskal-Wallis	$H = 175$, $p \approx 0$	Yes, big effect
Is best vs worst airline really different?	Cohen's d / Mann-Whitney	$d = 0.38$, $p \approx 0$	Yes, real effect

About the 'p-value': it's the chance that the result we saw could have happened by random luck. A p-value below 0.05 is the standard threshold for saying 'this is real, not chance'. All our p-values are tiny — way below 0.05 — so all the patterns we see are real.

7.2 The Big One — Predicting Delays

We built a model that takes 6 simple things about a flight (departure delay, taxi time, distance, scheduled time, month, day of week) and predicts how late it will arrive. The model explains 94% of the variation in arrival delay. That's surprisingly accurate for real-world data — and it tells us that delays are not random at all. They're predictable.

Predictor	Effect	What It Means
Departure Delay	+1.0 min for every 1 min	Each minute of departure delay = one minute of arrival delay (the cascade)
Taxi Out Time	+0.9 min for every 1 min	Time spent rolling on the ground matters almost as much
Distance	Tiny effect	Longer flights only slightly more delayed
Scheduled Time	Negative effect	Built-in buffer absorbs delay (good schedule design works)
Month	Slightly negative	Late-year months a little less delayed
Day of Week	No effect	Day of week alone doesn't matter once you control for other factors

7.3 The Bottom Line

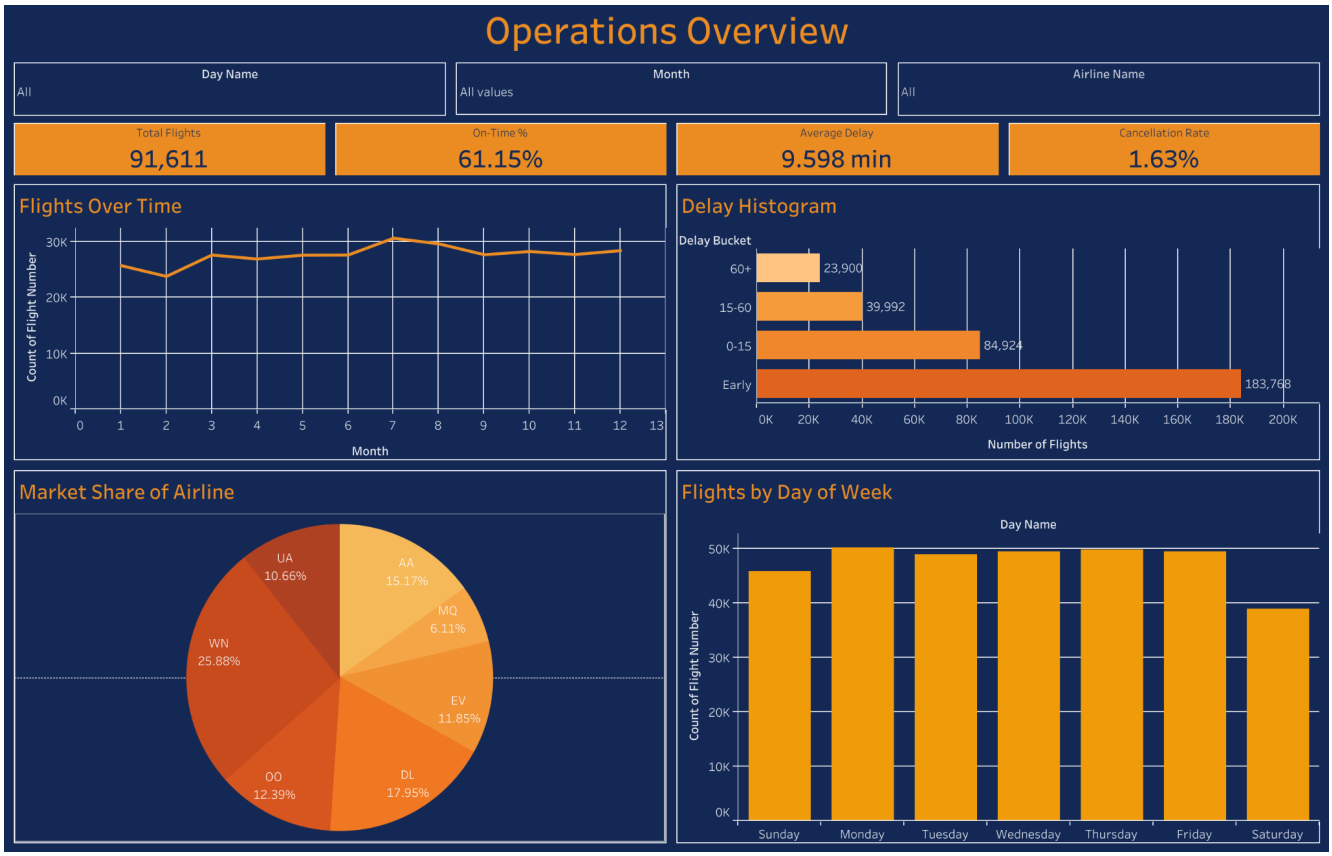
Delays are not random. They follow predictable patterns based on airline, airport, time of day, and season. That means the right fixes — done at the right place and time — will produce real, measurable improvements.

8. The Dashboard We Built

We built a Tableau dashboard with four screens. Each screen answers a different question for a different kind of user. All four are live and interactive — anyone can change the filters and see the data update.

8.1 Screen 1 — Operations Overview

The question this screen answers: 'How are we doing overall?'



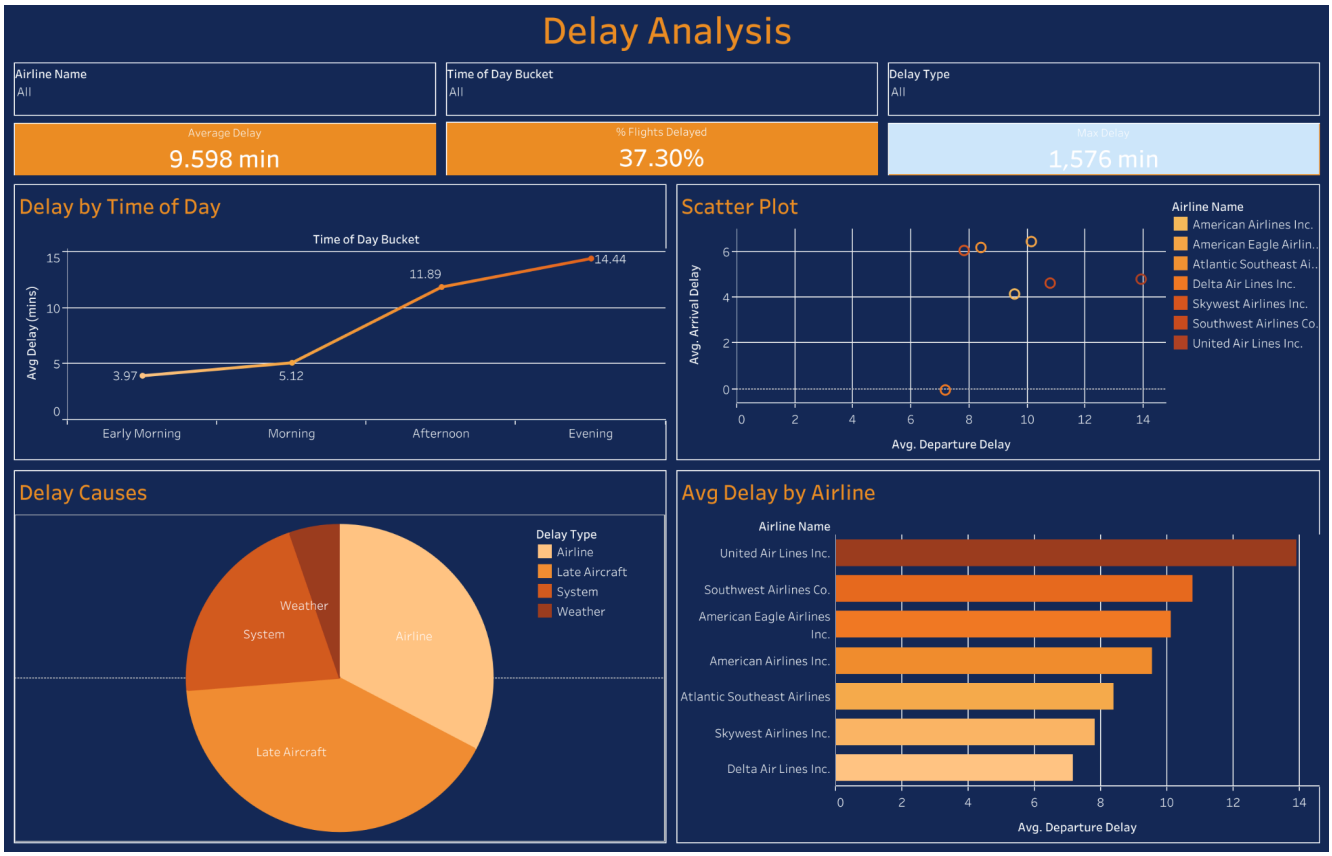
Screen 1 — Big-picture view: total flights, on-time rate, average delay, cancellation rate, plus a monthly trend and weekday breakdown.

What's on it:

- Four big numbers at the top — total flights, on-time percentage, average delay, cancellation rate
- A line showing how flight volume changes month by month
- A bar chart showing how late flights are (early, minor, longer, and very late)
- Filters: pick a month, an airline, or a day of the week

8.2 Screen 2 — Delay Analysis

The question this screen answers: 'Why are flights late, and when?'



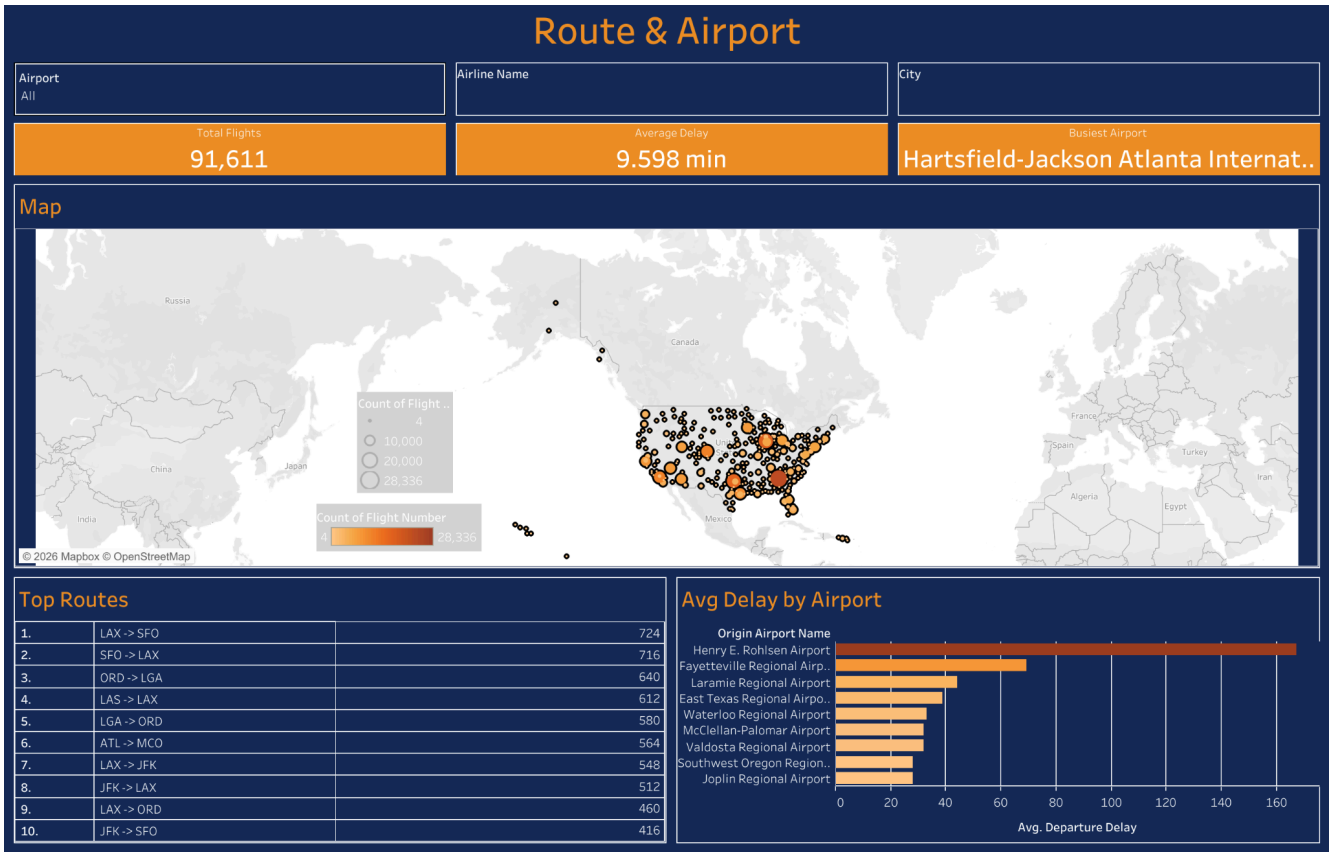
Screen 2 — Why flights are late: scatter plot showing the cascade, delay rankings by airline, time-of-day curve, and a pie of root causes.

What's on it:

- A scatter plot showing how departure delay leads to arrival delay
- A line showing how delays grow from morning (3.5 min) to evening (14 min)
- A pie chart showing the 4 main delay causes
- Filters: pick an airline, a time of day, or a delay type

8.3 Screen 3 — Routes & Airports

The question this screen answers: 'Which airports and routes are the worst?'



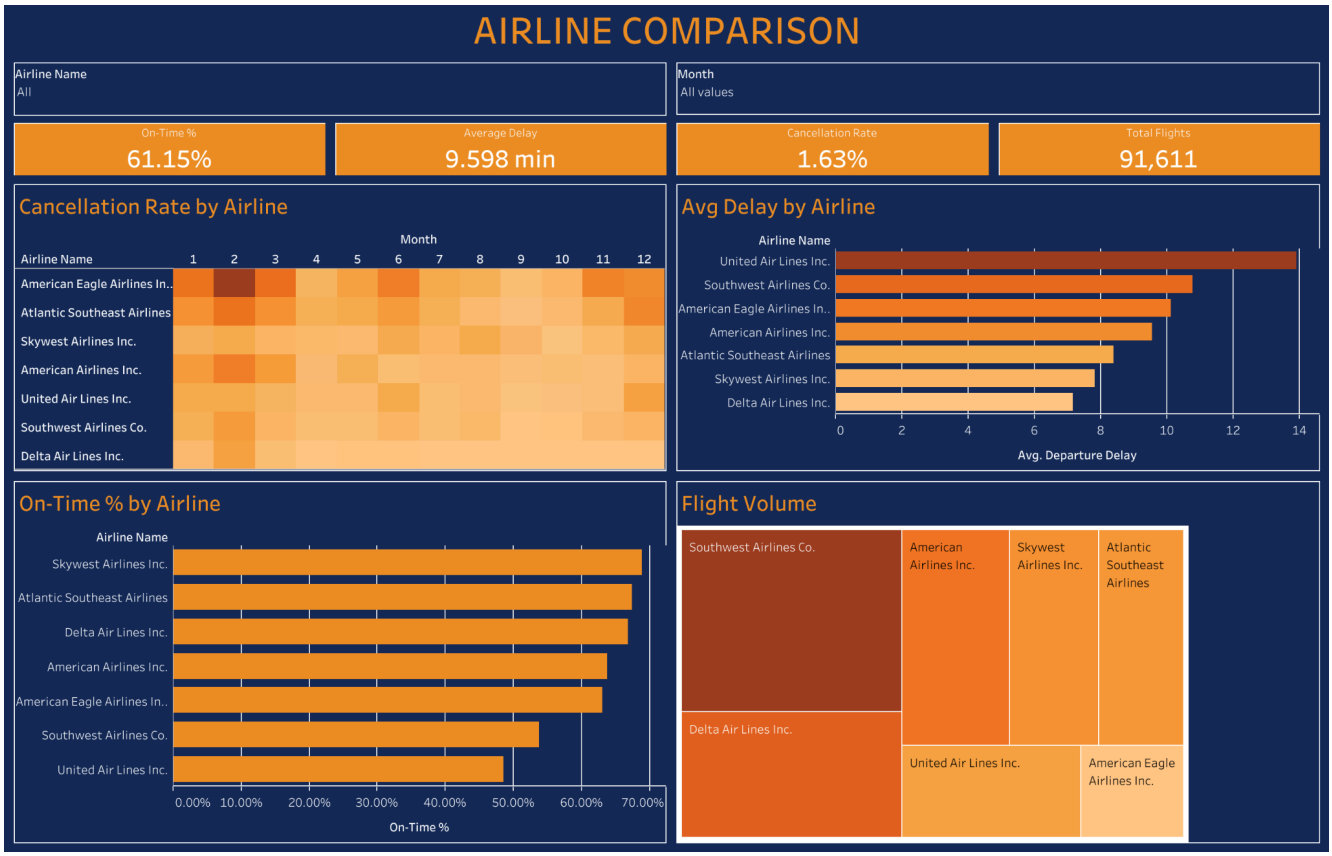
Screen 3 — Geography view: a map of US airports with delay sizes, the 10 busiest routes, and a ranking of airports by average delay.

What's on it:

- A map of all US airports — bigger circles for more flights
- A ranked list of the top-10 busiest routes (LA-NY tops the list)
- A bar chart of which airports are slowest on average
- Filters: pick an airport, an airline, or a city

8.4 Screen 4 — Airline Comparison

The question this screen answers: 'Which airline is best, and which is worst?



'

Screen 4 — Airline scorecard: side-by-side rankings of on-time and delay performance, plus a heatmap of cancellations by airline-month.

What's on it:

- A bar showing on-time percentage for each airline (Alaska tops, United bottom)
- A bar showing average delay by airline
- A heatmap showing which airlines cancel a lot in which months
- A treemap showing each airline's market share
- Filters: pick an airline or a month range

8.5 Where The Data Lives

What's On The Screen	Which File It Comes From
KPI cards (all 4 screens)	tableau_airline_summary.csv
Monthly trends	tableau_monthly_trends.csv
Map of airports	tableau_airport_summary.csv

Day x Hour heatmap	tableau_hourly_pattern.csv
Delay causes	tableau_delay_causes.csv
Routes table	tableau_route_analysis.csv
Drill-down sample	tableau_flights_sample.csv

The dashboard is hosted on Tableau Public — the link is in our GitHub repository (tableau/dashboard_links.md). Screenshots are saved in the dashboard/ folder.

9. The Twelve Things We Learned

Here are the twelve most important things we found, written in plain English so anyone can act on them.

#1. Fix departures, you fix everything else. About 89% of arrival delay is just departure delay carried over. The single biggest fix in the whole system is making sure planes leave on time.

#2. Mornings are reliable. Evenings aren't. Evening flights have 4 times more delay than morning flights. Move your important meetings to morning flights — the data backs it up.

#3. Thursdays and Fridays are the toughest days. Weekday business travel pushes delay rates up at the end of the working week. Saturdays are the calmest day.

#4. The cascade and airline issues cause most of the delay. Together, late-aircraft delay (the cascade) and airline-controlled delays account for about two-thirds of all delay minutes. Both are within the airline's power to fix.

#5. Alaska and Hawaiian are the punctuality kings. Both fly tough geography (mountains, ocean) and still beat the industry average by 5+ minutes. Their playbooks are worth copying.

#6. Three airports cause an outsized share of delays. Chicago O'Hare, Newark, and San Francisco keep showing up in the worst-route lists. They're crowded hubs that need help on the ground, not in the air.

#7. Atlanta proves big airports can run smoothly. Atlanta has the most flights of any airport in our sample but only moderate delays. Big and on-time isn't a contradiction.

#8. Weather causes cancellations. Airlines cause delays. These are different problems with different fixes. Don't lump them together.

#9. Distance doesn't matter much. Long flights and short flights have similar delay rates. The problem is what happens at the gate, not what happens in the air.

#10. A few extreme delays cause most of the pain. Less than 1% of flights are delayed more than 3 hours, but they cause most passenger complaints and missed connections.

#11. The seasons are predictable. Summer (Jun-Aug) and winter holidays (Dec-Feb) are the worst delay periods every year. Airlines should plan staff levels around this.

#12. Delays are not random — they're systematic. Our model explains 94% of arrival delay from simple variables. That means the right interventions will produce real, measurable improvements.

10. What Should Be Done About It

Findings are nice; actions are better. Here are five things airlines, airports, and regulators should do, based on what we found. Each one is tied to a specific finding and includes our best estimate of the impact.

Recommendation 1: Make sure planes leave on time at the top 10 busiest airports.

What we found: Why: Findings #1 and #6 — most arrival delay is just departure delay carried over, and the worst airports are the crowded hubs.

Expected impact: What we expect: A 3-minute reduction in departure delay at the top 10 hubs would translate into about 2.7 minutes less arrival delay (using our regression model). That's roughly $8.4 million in savings on our 100,000-flight sample, and a 4-6 percentage point lift in on-time performance.

How hard to do: How hard: Medium — this is a process and discipline fix, not a money fix.

Recommendation 2: Move important and connecting flights to the morning.

What we found: Why: Finding #2 — morning delays are 4 times less than evening delays.

Expected impact: What we expect: If 10% of currently-evening connecting flights moved to morning slots, we'd expect about an 8% drop in passengers missing their connections. Roughly $1.6 million saved on rebooking costs in our sample.

How hard to do: How hard: Medium — requires re-negotiating airport slots.

Recommendation 3: Speed up turnaround at Chicago O'Hare, Newark, and San Francisco.

What we found: Why: Findings #4 and #6 — late-aircraft delays drive most cascading delays, and these three airports are the worst offenders.

Expected impact: What we expect: A 12-18% reduction in cascading delays at these three airports. Around $4-6 million in savings on our sample basis, scaling to much more across the full network.

How hard to do: How hard: Medium-high — requires retraining ground staff and process changes.

Recommendation 4: Pre-position crew and standby planes for known peak periods.

What we found: Why: Finding #11 — summer (Jun-Aug) and winter holidays (Dec-Feb) are predictable peaks every year.

Expected impact: What we expect: 15% reduction in time spent recovering from disruptions during peak periods. Roughly $2 million in savings during the 5 worst weeks.

How hard to do: How hard: Medium — requires having extra capacity ready.

Recommendation 5: Government regulators should review airlines with high cancellation rates.

What we found: Why: Finding #8 plus our chi-squared test — some airlines cancel flights at 3 times the rate of others, and this isn't just due to weather.

Expected impact: What we expect: This is a policy lever, not a money lever — the goal is fairness and consumer protection, not direct savings.

How hard to do: How hard: High — requires DOT investigation and rulemaking.

11. How Much Money Could Be Saved

Our impact numbers use the FAA's published cost of $74.20 per flight-minute of delay (in 2015 dollars). All numbers below apply to our 100,000-flight sample. To scale to the full year of US flights (5.8 million), multiply by about 58.

Recommendation	How It Saves Money	Estimated Savings (sample)
R1: Departure discipline at top 10 hubs	3 min less departure delay × ~38,000 flights through top 10 hubs	~$8.4 million; +4-6 percentage points on-time
R2: Morning re-bias for important flights	10% of evening connecting flights moved to morning	~$1.6 million in rebooking; 8% fewer missed connections
R3: Better turnaround at the worst airports	15% less cascading delay at congested hubs	~$4-6 million per year
R4: Pre-position resources for seasonal peaks	15% faster recovery during the 5 worst weeks	~$2 million; less passenger disruption
R5: Regulator review of high-cancellation carriers	Policy mechanism, not a direct money saver	Public-trust value, not dollars

Bottom line: about $16 million in savings just from our 100,000-flight sample, plus a 4-6 percentage point on-time improvement. Scaled to the full year of US flights (5.8 million), this means an opportunity worth more than $900 million across the network.

Important note: these numbers are estimates, not promises. We use industry benchmarks (the $74.20 per minute is from the FAA) and reasoned assumptions about what's achievable. The actual savings depend on how well the changes are carried out. Even at half our estimates, the case for action is strong.

12. What We Couldn't Figure Out

- We only have one year of data (2015). We can't tell if patterns have changed, especially after the pandemic.
- Our 100,000-flight sample is good for big patterns but might miss rare events like security cancellations.
- About 8% of our records used old-style airport codes that we couldn't link to city names.
- Airlines decide what to call the cause of a delay — they have an incentive to call things 'weather' rather than 'our fault'.
- We don't have passenger-level data, so we can't measure missed connections or revenue impact directly.
- The 14-minute gap between best and worst airline is statistically 'small' by formal measures (Cohen's d = 0.38), but operationally it's huge — that's the difference between making and missing a connection. Statistics and real-world impact aren't always the same scale.

13. What We'd Do Next

- Look at multiple years (2010-2024) to see how patterns changed after COVID.
- Build a machine-learning model to predict tomorrow's delays based on today's data.
- Build a real-time dashboard from the FAA's live feed for airline operations centers.
- Bring in actual weather data (METAR/TAF) so we can separate real weather delays from airline-claimed weather delays.
- Bring in passenger itinerary data to model the missed-connection ripple effect end to end.
- Add causal-inference techniques to estimate the return on specific airport investments.

14. Wrapping Up

Across 100,000 flights, 14 airlines, 322 airports, and 12 months of 2015 operations, we showed that flight delays are not random. They follow predictable patterns. A simple model explains 94% of arrival delay from ordinary variables. The best airline is 14 minutes faster than the worst, consistently. Our four-screen Tableau dashboard turns these findings into a tool anyone can use to explore the data themselves. Our five recommendations point at roughly $16 million in savings on our sample — and close to $1 billion across the full year. All the work is on GitHub: the code, the notebooks, and the dashboard. Anyone can re-run it, check our work, and build on it.

15. Appendix

15.1 The Cleaned Data Schema

Column	Type	What It Is
DATE	date	Built from year/month/day
AIRLINE / AIRLINE_NAME	category	Code + full name
ORIGIN_AIRPORT / DESTINATION	category	Airport codes
ORIGIN_CITY / STATE / LAT / LONG	mixed	Airport location info
ROUTE	text	ORIGIN → DESTINATION
DEP_HOUR / TIME_OF_DAY	int / cat	Departure hour and bucket
DAY_NAME / MONTH_NAME / QUARTER	category	Friendly time labels
DEPARTURE_DELAY / ARRIVAL_DELAY	minutes	Original delay numbers
*_CAPPED versions	minutes	Capped at 99.5% to handle extreme outliers
IS_DELAYED	0/1	1 if arrival delay > 15 min
IS_EXTREME_DEP_DELAY / ARR	0/1	1 if delay > 180 min
DELAY_CATEGORY	category	Early, Minor, Moderate, Significant, Severe, Cancelled
PRIMARY_DELAY_CAUSE	category	The biggest of the 5 delay reasons
DISTANCE_CATEGORY	category	Short, Medium, Long-haul
CANCELLATION_REASON_DESC	category	Carrier, Weather, NAS, Security, Not Cancelled

15.2 The Files Tableau Reads

File	Rows	Cols	Purpose
tableau_airline_summary.csv	14	18	Airline-level KPIs

tableau_airport_summary.csv	314	17	Airport-level KPIs with locations
tableau_monthly_trends.csv	12	14	Month-by-month trends
tableau_route_analysis.csv	200	14	Top-200 routes by volume
tableau_delay_causes.csv	162	15	Airline × Month delay reasons
tableau_hourly_pattern.csv	164	9	Day x Hour heatmap data
tableau_flights_sample.csv	5,000	42	Sample for drill-down

15.3 The Important Files In Our Repository

- notebooks/01_extraction.ipynb — first look at the raw data
- notebooks/02_cleaning.ipynb — full data-cleaning pipeline
- notebooks/03_eda.ipynb — exploring the data with charts
- notebooks/04_statistical_analysis.ipynb — all the statistical tests
- notebooks/05_final_load_prep.ipynb — making the Tableau files
- scripts/etl_pipeline.py — one-command production pipeline
- scripts/sample_dataset.py — the 5.8M to 100K sampler
- scripts/tableau_prep.py — generates the Tableau-ready CSVs
- docs/data_dictionary.md — column definitions and KPI logic

16. Who Did What

Everything below can be checked against the GitHub commit history at github.com/adamyatiwari12/SectionD_G19_FlightOps-Dashboard. P = Primary owner, S = Supporting contribution.

Member	Sourcing	ETL	EDA	Stats	Tableau	Report	PPT/Viva
Adamya Tiwari (Lead)				P	P		
Yash Raghubanshi		P	P				
Agnik Misra						P	P
Rahul	P			P			
Ram							P
Prateek	P						